

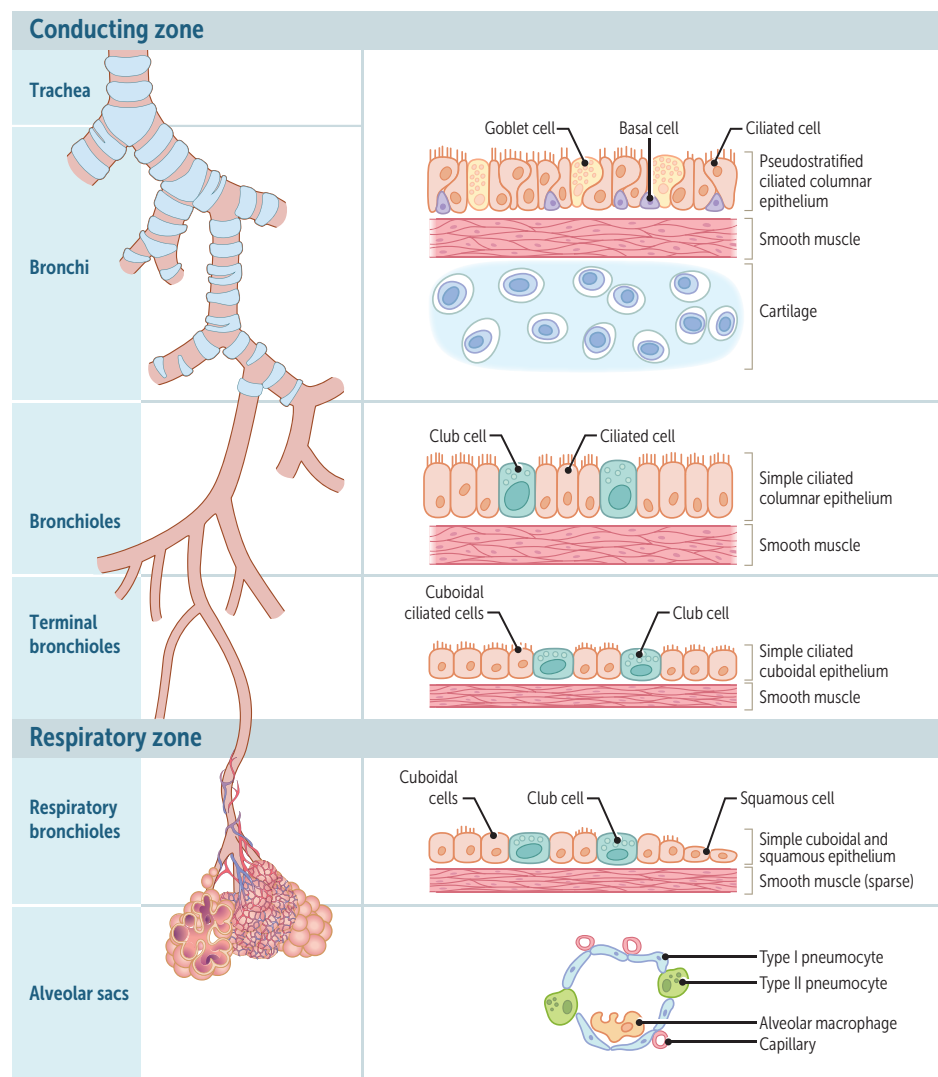
## ► RESPIRATORY—ANATOMY

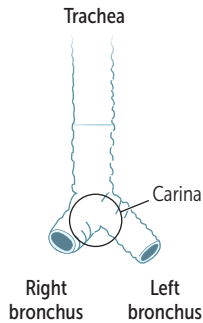
**Respiratory tree****Conducting zone**

Large airways consist of nose, pharynx, larynx, trachea, and bronchi. Airway resistance highest in the large- to medium-sized bronchi. Small airways consist of bronchioles that further divide into terminal bronchioles (large numbers in parallel → least airway resistance).  
 Warms, humidifies, and filters air but does not participate in gas exchange → “anatomic dead space.”  
 Cartilage and goblet cells extend to the end of bronchi.  
 Pseudostratified ciliated columnar cells primarily make up epithelium of bronchus and extend to beginning of terminal bronchioles, then transition to cuboidal cells. Clear mucus and debris from lungs (mucociliary escalator).  
 Airway smooth muscle cells extend to end of terminal bronchioles (sparse beyond this point).

**Respiratory zone**

Lung parenchyma; consists of respiratory bronchioles, alveolar ducts, and alveoli. Participates in gas exchange.  
 Mostly cuboidal cells in respiratory bronchioles, then simple squamous cells up to alveoli. Cilia terminate in respiratory bronchioles. Alveolar macrophages clear debris and participate in immune response.



**Lung anatomy**

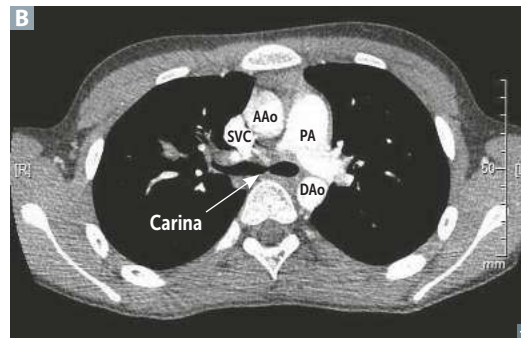
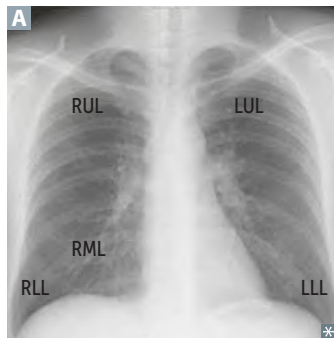
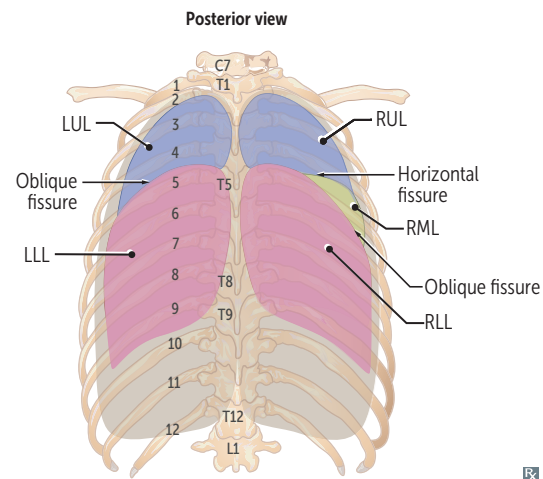
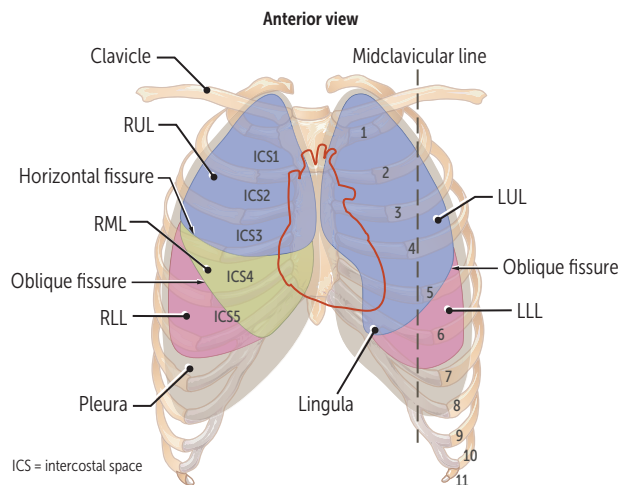
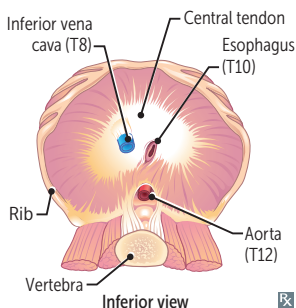
Right lung has 3 lobes; Left has less lobes (2) and lingula (homolog of right middle lobe). Instead of a middle lobe, left lung has a space occupied by the heart **A**.

Relation of the pulmonary artery to the bronchus at each lung hilum is described by **RALS**—Right Anterior; Left Superior. Carina is posterior to ascending aorta and anteromedial to descending aorta **B**.

Right lung is a more common site for inhaled foreign bodies because right main stem bronchus is wider, more vertical, and shorter than the left. If you aspirate a peanut:

- While supine—usually enters superior segment of right lower lobe or sometimes enters posterior segment of right upper lobe.
- While lying on right side or prone—usually enters right upper lobe.
- While upright—usually enters right lower lobe.

**Thoracentesis**—pleural space between visceral and parietal pleura. Thoracentesis → remove liquid or air from this space. Needle above the rib → avoid neurovascular bundle below each rib. Avoid below 9th rib → risk of injuring abdominal structures (ie, right hepatic lobe, spleen).

**Diaphragm structures**

Structures perforating diaphragm:

- At T8: IVC, right phrenic nerve
- At T10: esophagus, vagus (CN 10; 2 trunks)
- At T12: aorta (red), thoracic duct (white), azygos vein (blue) (“At T-1-2 it’s the red, white, and blue”)

Diaphragm innervated by C3-5 (phrenic). Pain from diaphragm irritation can be referred to shoulder (C5) and trapezius ridge (C3, 4). Phrenic nerve injury causes elevation of the ipsilateral hemidiaphragm on x-ray.

Number of letters = T level:

T8: vena cava (IVC)

T10: (O)esophagus

T12: aortic hiatus

I ate (8) ten eggs at twelve.

C3, 4, 5 keeps the diaphragm alive.

Other bifurcations:

- The Common Carotid bifurcates at C4.
- The Trachea bifurcates at T4.
- The abdominal aorta bifurcates at L4.